Confidently Wrong: Ranking Inversion in Cross-Network Denial-of-Service Detection

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Abstract

Flow-based DDoS detectors report accuracies above 99% and lose most of that skill when applied to a different network. The loss is well documented; its character is not, and the difference decides which remedies can work. Training four model families on each of four independently constructed testbeds under a published common feature set, and evaluating every ordered pair, we find that cross-network transfer does not decay toward chance. It passes through it. Pooling four model families, twelve ordered pairs and five seeds, 108 of 240 external evaluations score below 0.5 and mean external ROC-AUC is 0.5054; and one pair reaches 0.0697, meaning attack flows are systematically assigned lower scores than benign flows; negating that detector's score would yield an ROC-AUC of approximately 0.9303. A model at chance carries no usable ordering; a model at 0.0697 carries a strong one with the sign reversed. Three consequences follow. Feature engineering along the obvious semantic axis does not help: we pre-specified the hypothesis that capacity-bound features carry testbed identity, and removing all twenty of them changes external macro-F1 by +0.0132 [-0.0084, +0.0320], an interval spanning zero, so the hypothesis is not supported at the power this study affords. A one-feature set does reach external macro-F1 0.6512 against 0.3748 for all 39, a paired difference of +0.2763 [+0.2230, +0.3366], but a control over random subsets at controlled size shows that dimensionality accounts for the smaller part of this and the identity of that feature — a DNS indicator that is itself a dataset shortcut — for the larger. Feature count tracks in-domain separability (Spearman +0.626) and external macro-F1 (-0.290) while leaving external ranking untouched (+0.004 [-0.081, +0.088]), making it a threshold effect rather than a representational one. Distribution-divergence monitoring did not work in this benchmark: proxy A-distance, RBF-MMD and summed Wasserstein distance all predicted the observed loss worse than assuming the average drop, while the model's own confidence (ATC) reaches Spearman -0.726 and cuts leave-one-pair-out error by 24%. Recalibration works for one pair in twelve: in the primary LightGBM analysis it is the only pair whose ranking survived, and there a threshold read from target benign traffic recovers 97.0% of the available headroom using no attack labels, but pooled over held-out evaluation sets that recalibration is net harmful. Ten labelled target flows, used to retrain, reach median macro-F1 0.7253 [0.5604, 0.9540] and exceed what the best achievable threshold reaches with unlimited target labels (0.6158). Finally, the failure mode depends strongly on the dataset pair: the same-lineage CIC pairing failed mainly at the threshold and looked recoverable, while the independently built transfers failed at the ranking. Cross-network evaluation therefore has to distinguish ranking failure from threshold failure, because the two admit different remedies.

Keywords: network intrusion detection, DDoS, cross-dataset generalisation, distribution shift, calibration, evaluation methodology

1 Introduction

Published flow-based DDoS detectors report accuracies above 99%. Deployed ones do not behave that way. The gap between those two facts has been measured repeatedly: models trained on one intrusion-

detection dataset and tested on another lose most of their apparent skill [D'hooge 2020; Cantone 2024]. That finding is established. What it *means* is not, and the ambiguity is expensive, because the two available readings prescribe opposite engineering.

The default reading is quantitative. The model is somewhat overfitted to its training network and performance degrades, so the remedies are the familiar three: engineer features that describe the attack rather than the capture, monitor the distance between training and deployment distributions, and recalibrate the decision threshold on arrival. This paper tests a representative version of each and finds the reading incomplete in a way that matters.

There is a structural reason to expect the two readings to diverge. Drift monitoring watches $P(X)$, the distribution of the features. What decides whether a detector still works is $P(Y \mid X)$, the relationship between features and labels. Nothing ties the two together: a target network can look statistically familiar and score differently, or look alien and score the same. That gap is what the experiments below exploit, and it is why the divergence measures in §4.7 fail as risk predictors while a measure of the model's own behaviour does not.

Running every ordered pair among four independently captured testbeds, on the NetFlow v2 common feature set, with exact de-duplication and source-only preprocessing, we find that transfer does not decay toward chance. It passes through chance. Mean external ROC-AUC is 0.5054; 108 of 240 run-level external evaluations fall *below* 0.5, as do 5 of the 12 ordered pairs once seeds are averaged; one pair reaches **0.0697**.

That last number is what an aggregate near chance conceals. An AUC of 0.0697 is not weak evidence; it is strong evidence with the wrong sign. Attack flows are systematically assigned lower scores than benign flows, and negating the detector's output would yield an ROC-AUC of approximately 0.9303. Note that this is a statement about ranking, not about accuracy: ROC-AUC is threshold-free, and the accuracy of a negated detector would still depend on the operating point and the target prevalence. A model at chance carries no usable ordering. A model at 0.0697 carries a strong ordering with the sign reversed.

A plausible mechanism is easy to state. If high throughput marks attack traffic in one testbed and marks the busiest legitimate hosts in another, a decision surface fitted to the first is close to optimal-but-negated on the second. Two of our own results constrain what that mechanism can be. Logistic regression inverts on the same pairs as the tree ensembles (§4.2), so it cannot require high-order feature interactions, which a linear model cannot represent. And feature count is uncorrelated with external ranking quality (§4.4), so it is not a matter of model capacity either. What remains is the simplest reading: individual feature-label associations reverse sign between these networks. We do not isolate which associations, and §6 says so.

Four consequences follow. The first three narrow a standard remedy; the fourth conditions all of them:

Feature engineering along the obvious axis does not help. We pre-specified the expectation that capacity-bound features (throughput, duration, volume) carry testbed identity and cause the collapse, and committed it to the project repository before the experiment ran. Removing all twenty of them moves external macro-F1 by +0.0132 [-0.0084, +0.0320], an interval spanning zero, and capacity features alone transfer as well as the full set. The hypothesis is not supported. A one-feature set does out-transfer the full set by +0.2763 [+0.2230, +0.3366], but a control over randomly chosen subsets at controlled size shows dimensionality to be the smaller part of that advantage. Random subsets gain +0.105 on the way from 39 features down to one, while the chosen feature beats a random single feature by +0.147. The two are measured against different baselines and are not an additive split of +0.2763 (§4.4); what they jointly establish is that most of the advantage is the feature, which turns out to be a second dataset shortcut.

Distribution-divergence monitoring did not work here. Proxy A-distance, RBF-MMD and summed Wasserstein distance, the quantities motivating much of the domain-adaptation literature, all have correlations whose confidence intervals span zero, and all three predict the observed loss *worse* than assuming the average drop. What does carry signal is the model's own confidence: Average Thresholded Confidence reaches Spearman -0.726 and cuts leave-one-pair-out error by 24%.

Recalibration works for one pair in twelve. Decomposing each failure in the primary LightGBM analysis into ranking loss and threshold loss shows eleven of twelve fail representationally. For the single ranking-preserving pair, a threshold read from target benign traffic, which requires no attack labels, recovers 97.0% of the available headroom. On three others it is actively harmful. Ten labelled target flows, drawn from an adaptation pool disjoint from the evaluation set, lift median external macro-F1 from 0.3919 to 0.7253 and beat the best threshold obtainable with unlimited labels.

Dataset quality bounds what any of this measures. In the feature space a model actually sees, these datasets are up to 99.7% exact duplicates: NF-BoT-IoT's 30,420,086 rows reduce to 101,369. A random split that retains these duplicates places indistinguishable feature vectors on both sides of it, so the resulting scores are not independent test evidence.

1.1 Contributions

1. Evidence that cross-network failure can manifest as systematic **ranking inversion** rather than monotonic degradation, across 240 model-pair-seed evaluations, with permutation and label-reversal controls establishing that the below-chance results are not a pipeline artefact.
2. A pre-specified capacity-feature hypothesis that the data does not support, reported with its interval, alongside a controlled random-subset analysis showing that feature-count effects act on the operating point rather than on external ranking.
3. A label-free triage analysis in which Average Thresholded Confidence beats the mean-gap baseline under leave-one-pair-out evaluation while three marginal distribution-distance measures do not.
4. A failure taxonomy separating threshold failure from ranking failure, and ranking failure from sign inversion, together with evidence that which mode appears depends strongly on the source-target pairing.
5. A benchmark-integrity audit covering exact duplication, contradictory labels under the full attack taxonomy, non-finite extractor artefacts and corrected labels, showing that each can materially change measured transfer performance.

2 Related work

2.1 The cross-dataset gap

That flow-based intrusion detectors transfer badly is not in dispute. D'Hooge et al. [2020] established reference scores on CIC-IDS2017 and CSE-CIC-IDS2018 across twelve supervised algorithms and concluded that the classifiers could not operate reliably on traffic from a network unseen at training time. Cantone et al. [2024] extended the design to four datasets and four classifiers, reported cross-dataset accuracy largely commensurate with random chance, and traced part of it to anomalies in the data itself. The question has stayed open: Hossain et al. [2026] report cross-dataset accuracy below 40% between UNSW-NB15 and ToN-IoT for models scoring above 95% within either one, and Elangovan et al. [2026] find the same pattern across four datasets when generalisation is tested temporally and semantically rather than within a single capture.

Our study shares their protocol and departs on the question. Prior work establishes *that* the gap exists and how large it is. We ask what kind of failure it is. That distinction is not rhetorical: an aggregate accuracy near chance is compatible with a detector that has lost its signal and with one that has reversed it, and only the second is visible once ROC-AUC is reported per ordered pair. The remedies implied by the two readings differ completely, which is why we test a representative version of each rather than proposing a fourth.

2.2 Feature standardisation and dataset quality

Sarhan et al. [2021] identified the absence of a common feature set as a structural obstacle to cross-dataset claims, and released NetFlow v2 conversions of four datasets sharing 43 fields. We adopt that feature set precisely so our transfer results cannot be attributed to ad-hoc harmonisation, and verify the shared space by assertion rather than assumption. The four source datasets themselves are CSE-CIC-IDS2018 [Sharafaldin et al. 2018], BoT-IoT [Koroniotis et al. 2019], ToN-IoT [Alsaedi et al. 2020] and UNSW-NB15 [Moustafa and Slay 2015]; the secondary study uses CIC-DDoS2019 [Sharafaldin et al. 2019] and CICIDS2017 [Sharafaldin et al. 2018].

A second line of work asks not whether the labels are right but whether the features are honest. D'hooge et al. [2022a] showed that metadata fields, destination port above all, act as shortcuts strong enough to separate ten intrusion-detection datasets at 70–100% accuracy on their own; D'hooge et al. [2022b] extended the search to non-metadata contaminants, and D'hooge et al. [2021] showed how much of the apparent value of a large feature set is carried by a small block of it. Goldschmidt and Chudá [2025] survey 89 public NIDS datasets and reach a compatible conclusion about how dataset construction shapes what models learn. Our `DNS_QUERY_ID` result (§4.4) is an instance of the phenomenon they describe, and we do not present shortcut features as a discovery. What is new here is what a shortcut does across networks rather than within one: the same feature space that inflates in-domain scores can also produce a *reversed* target ordering, and the failure mode determines whether recalibration or target supervision is the right response.

Dataset quality has its own literature. Engelen et al. [2021] documented labelling and traffic-generation defects in CICIDS2017 and released a corrected version; Lanvin et al. [2022] extended the analysis and quantified the effect of the corrections on measured performance. Both studies examine in-domain consequences. We use the corrected release differently, as an instrument: applying one identical source model to the original and corrected targets isolates label error from domain shift, and the result runs opposite to the intuitive direction (§4.11).

Our duplicate accounting (§3.3) is adjacent to this work but, to our knowledge, has not been reported at this scale for the NetFlow v2 datasets. That three of four are more than 80% duplicate in the model's own feature space constrains how in-domain numbers computed on them should be read, and §4.6 measures how much it moves the transfer results themselves.

2.3 Methodology and evaluation norms

Arp et al. [2022] catalogue ten recurring pitfalls in machine learning for security, of which sampling bias, data snooping, spurious correlations and the base-rate fallacy bear directly on cross-dataset evaluation. We map the protocol against all ten in Table 2 rather than asserting compliance in prose. Axelsson [2000] established that intrusion detection is dominated by the base rate; the datasets here span base rates from 1.3% to 77.4%, so balanced-test-set F1 is not comparable across them and operating-point metrics are reported alongside.

2.4 Predicting performance without labels

Estimating target accuracy from unlabelled data is an active area outside security. Garg et al. [2022] introduced Average Thresholded Confidence, which learns a confidence threshold on source data and predicts accuracy as the fraction of unlabelled target points above it. The competing family of approaches measures distance between source and target feature distributions directly, which is also the theoretical motivation for domain-adversarial training. Section 4.5 evaluates both families on the same twelve ordered pairs and finds their ordering reversed relative to the assumption embedded in most of that work: the distribution-distance measures lose to a mean baseline, and the confidence-based estimator does not.

2.5 Domain adaptation for intrusion detection

Adapting a detector to a new network is a well-populated area, including covariance alignment [Sun et al. 2016] and domain-invariant representation learning for NIDS specifically. We make no methodological contribution here and do not claim one. CORAL appears in §4.8 as the standard zero-label baseline, and the contribution is the measurement: what the available zero-label methods actually recover in this setting, which is close to nothing, against what a handful of labelled target flows recovers, which is most of the gap.

3 Data and experimental protocol

The design in one paragraph. Four intrusion-detection datasets, each captured on a different testbed, are reduced to a single shared 39-feature space. For each dataset in turn, four model families are trained on 80% of it and evaluated twice: on its own held-out 20%, which we call **in-domain**, and on each of the other three datasets, which we call **external**. In the primary transfer experiment nothing about the target is used in fitting: not its labels, not its feature statistics, not its threshold. The quantity of interest is the difference between those two evaluations, and every design decision below exists to ensure that difference measures domain change rather than an artefact of preparation.

3.1 A published feature space, verified

Cross-dataset claims in this area usually rest on intersecting the column names of two CICFlowMeter exports. The resulting space is undocumented, varies between studies, and is itself a confound: a transfer result obtained this way cannot be separated from the harmonisation that produced it.

We use the NetFlow v2 feature set of Sarhan et al. [2021], built for this purpose. Its four datasets ship an identical 43-column schema. We verify rather than assume this: the loader asserts that the numeric feature sets of all four are equal, not merely overlapping, and fails otherwise. From the 43 columns we drop the two label columns and `L4_SRC_PORT` and `L4_DST_PORT`, leaving **39 features**. Ports are dropped because in these testbeds a destination port frequently identifies the attack script rather than the service. That is the spurious-correlation failure Arp et al. describe.

The conventional CICFlowMeter pairing (CIC-DDoS2019 ↔ CICIDS2017) is retained as a secondary study, so the primary conclusions can be checked in the feature space the literature actually uses.

3.2 Labels

The positive class is any DoS or DDoS flow; the negative class is benign. Attacks that are neither, such as exploits, fuzzers, port scans and brute-force, are not silently discarded: the policy is an explicit parameter and every run records how many rows it moved, because discarding them changes the base rate and the base rate drives precision.

Labels also map to attack families (volumetric, reflection/amplification, low-rate application-layer) so transfer can be reported per family. Family assignment resolves the longest matching pattern across all families rather than the first family that matches: `DoS slowloris` contains both `dos` and `slowloris`, and a first-match rule files a low-rate attack as a flood. Ties break on a declared precedence, which places CIC-DDoS2019's `DrDoS_*` labels in reflection/amplification. The complete label → class → family map for all seven datasets is released.

3.3 De-duplication, and why it changes the picture

Exact duplicates are removed **in the feature space the model actually sees**, after socket columns are dropped. Two flows differing only in source port are one example to a model that never observes the port, and letting such a pair straddle a train/test split is leakage.

Table 1: Datasets after preparation.

dataset	raw rows	after de-duplication	duplicate rate	DDoS/DoS flows	base rate
NF-CSE-CIC-IDS2018	17,129,715	8,981,062	47.6%	453,237	0.0505
NF-BoT-IoT	30,420,086	101,369	**99.7%**	53,005	0.5229
NF-ToN-IoT	13,135,881	2,477,944	81.1%	529,884	0.2138
NF-UNSW-NB15	1,986,745	236,999	88.1%	3,075	0.0130
CIC-DDoS2019	431,371	418,884	2.9%	324,197	0.7740
CICIDS2017	2,313,810	2,231,536	3.6%	321,706	0.1442
CICIDS2017 (corrected)	1,787,358	1,747,255	2.2%	266,899	0.1528

Three of the four NetFlow datasets are more than 80% duplicate. We additionally count *contradictory groups*, identical feature vectors carrying opposite labels, at 1,940 in NF-CSE-CIC-IDS2018 and 858 in NF-UNSW-NB15. Those counts are taken over the full attack taxonomy. Under the DoS-only policy the primary analysis actually uses, both fall to zero: every contradictory group in these two datasets pairs a benign flow with a *non-DoS* attack, and those rows are removed before modelling. The ceiling is therefore real for a general intrusion detector on these captures and absent for the DoS detector studied here, which is verified rather than assumed in §4.6.

3.4 Non-finite and impossible values

The per-second rate features carry finite but physically impossible magnitudes: `SRC_TO_DST_SECOND_BYTES` reaches 8.9×10^{304} in NF-ToN-IoT and `DST_TO_SRC_SECOND_BYTES` reaches 2.3×10^{252}, produced by dividing byte counts by near-zero flow durations. This is the extraction artefact Engelen et al. document. Such cells describe the extractor, not the traffic. They are set to missing and counted (213 in NF-CSE-CIC-IDS2018, 216 in NF-ToN-IoT) rather than allowed to become silent infinities under a numeric cast.

3.5 Training

The order of operations is the substance of the protocol:

$$\text{de-duplicate} \rightarrow \text{split} \rightarrow \text{fit scaler on } \textbf{source-train only} \rightarrow \text{transform everything else}$$

Splits are stratified 80/20 and seeded. Scaler and median imputer are fitted on source-train alone and applied unchanged to the source test split and to every target. Target features are clipped to the range the scaler observed; the clipping is ablated in §4.6, not assumed. Imbalance is handled with balanced sample weights, never resampling, in all model fitting reported here; the label-budget experiment of §4.8 deliberately draws a balanced labelled adaptation set, which is a controlled supervision-budget intervention rather than a training-time resampling choice, since SMOTE applied before a split leaks synthetic neighbours across it. Models are LightGBM (primary), XGBoost, RandomForest and LogisticRegression at fixed hyperparameters, with no tuning against any target. Five seeds are used for the transfer matrix and three for the ablations and remedy experiments, which are correspondingly more expensive; the asymmetry is a cost decision and every reported interval states the number of pairs it rests on.

A stratified cap of 50,000 rows per class bounds cost, and it is applied to **both** the source and the target: external metrics are computed on a seeded stratified sample of the de-duplicated target, not on the target in full. Because the cap equalises the two classes it also changes the evaluated prevalence, which moves every prevalence-sensitive metric. This is why ROC-AUC, which is invariant to prevalence under a fixed score ordering, carries the paper's central claim, and macro-F1 and MCC are read alongside it rather than in place of it.

The order of operations that carries the leakage guarantee is: de-duplicate, then split, then fit. Exact duplicates are removed from a dataset before any partition is drawn, so no feature vector can appear on both sides of a split. The imputer and the scaler are fitted on the source training partition alone and applied unchanged to the source evaluation partition and to every target; the decision threshold is chosen on the source evaluation partition (§3.6). No target label, target feature statistic or target-derived threshold enters the primary transfer experiment. A pipeline self-check verifies the scaler property on every run: a scaler fitted on train maps train exactly into [0, 1] **and** lets some unseen rows fall outside it, which a scaler that had seen those rows cannot do.

3.6 Choosing the operating threshold

The decision threshold is chosen on the source evaluation split, never on target data, and taken as the midpoint of the widest tied-optimal plateau, over candidates placed between consecutive score levels.

This is not a detail. In-domain AUC here routinely reaches 1.0000, so a wide interval of thresholds is exactly optimal. Selecting by `argmax` returns whichever endpoint the grid visited first, and that endpoint sits flush against the score distribution, maximally fragile under any shift. In our own runs the naive choice inflated apparent threshold loss on one pair from +0.0128 to +0.1948, an artefact of the tie-break with no relation to domain shift.

3.7 Metrics

To quantify the gap itself rather than its endpoints, we resample both test sets jointly and report a **paired bootstrap** interval and p-value on the difference (2,000 resamples). Reporting one interval for the in-domain score and another for the external score says nothing about the interval of their difference.

To separate a representation that stopped working from an operating point that went stale, we decompose each gap into **ranking loss** (Δ ROC-AUC) and **threshold loss** (external macro-F1 at the target-optimal threshold minus at the source threshold). The two have different remedies, and aggregating them hides which failure occurred.

To ask whether a detector's probabilities remain usable, we report **ECE** (15 equal-width bins) and **Brier score** in both domains, because a detector deploys a threshold rather than a ranking.

To keep the base rate visible, we report **FPR at 95% TPR**, since balanced-test-set F1 flatters every detector in this literature.

4 Results

4.1 In-domain, the problem looks solved

On its own held-out split every source reaches what this literature routinely reports. LightGBM averages 0.9817 macro-F1 across the four sources (0.9546–1.0000) at ROC-AUC 0.9977–1.0000. The other three model families agree within a few points, and seed-to-seed standard deviations reach at most 0.0023. Nothing here suggests an unsolved problem.

Table 2: Arp et al. [2022] pitfalls and how this protocol addresses each.

Pitfall	Addressed by
Sampling bias	Four independently collected testbeds; every ordered pair in both directions; per-dataset base rates reported, not normalised away
Label inaccuracy	Full label→class map released; contradictory groups counted; label error isolated against the corrected CICIDS2017
Data snooping	De-duplicate before splitting; scaler, imputer and threshold fitted on source data only, verified by the pipeline self-check (§3.5)
Spurious correlations	Socket columns dropped; feature classes ablated; `DNS_QUERY_ID` examined directly rather than trusted as noise
Biased parameter selection	Fixed hyperparameters across all domains; no tuning against any target
Inappropriate baseline	Four model families including logistic regression; a one-feature set as minimal reference
Inappropriate performance measures	macro-F1, MCC, AUC, PR-AUC, ECE, Brier, FPR@95%TPR reported together
Base rate fallacy	Base rates span 1.3%–77.4% and are stated; FPR at 95% TPR reported in Table 5; the balanced evaluation cap and its effect on prevalence-sensitive metrics stated in §3.5
Lab-only evaluation	Acknowledged in §6; all four datasets are testbed captures, which is part of the finding
Inappropriate threat model	Scope stated as DoS/DDoS detection under network change, not adversarial evasion

Two features of these numbers matter for what follows. First, they are obtained *after* de-duplication, so they are not inflated by the train/test contamination described in §3.3; the uncorrected figures would be higher still. Second, ROC-AUC reaches 1.0000 on NF-CSE-CIC-IDS2018 to four decimal places, meaning the classes are perfectly separable in the source domain. Perfect in-domain separability is what makes the threshold plateau of §3.6 wide, and it is the first indication that the source decision rule may be exploiting testbed-associated structure rather than attack behaviour alone.

4.2 Out-of-domain, the failure has a sign

Applying the identical fitted model to another testbed across all 12 ordered pairs and 5 seeds, LightGBM averages 0.3788 external macro-F1 (0.1935–0.5957). The mean paired gap is 0.6029 and the paired bootstrap rejects a zero gap in **60 of 60** pair-seeds at $p < 0.05$ (Figure 1).

The informative statistic is external ROC-AUC. It averages 0.5054, which is chance. A mean at chance could describe a detector that has merely lost its signal. That is not what happened:

- **108 of 240** (model, pair, seed) runs score below 0.5;
- **149 of 240** have negative MCC;
- NF-UNSW-NB15 → NF-CSE-CIC-IDS2018 reaches **0.0697**, averaged over its five seeds.

Figure 2 shows the matrix. The entire NF-UNSW-NB15 source row is inverted against every target. This is the paper's central claim: cross-network failure can involve systematic ranking inversion rather than simple attenuation of the learned relationship. It is not the only mode observed. Of the 240 external evaluations, 108 fall below chance, so inversion is common rather than universal, and the taxonomy in §4.3 separates it from the milder modes.

The inversion is not a property of one learner. The same 12 ordered pairs and 5 seeds were run for four model families, which lets the question be asked directly rather than assumed away.

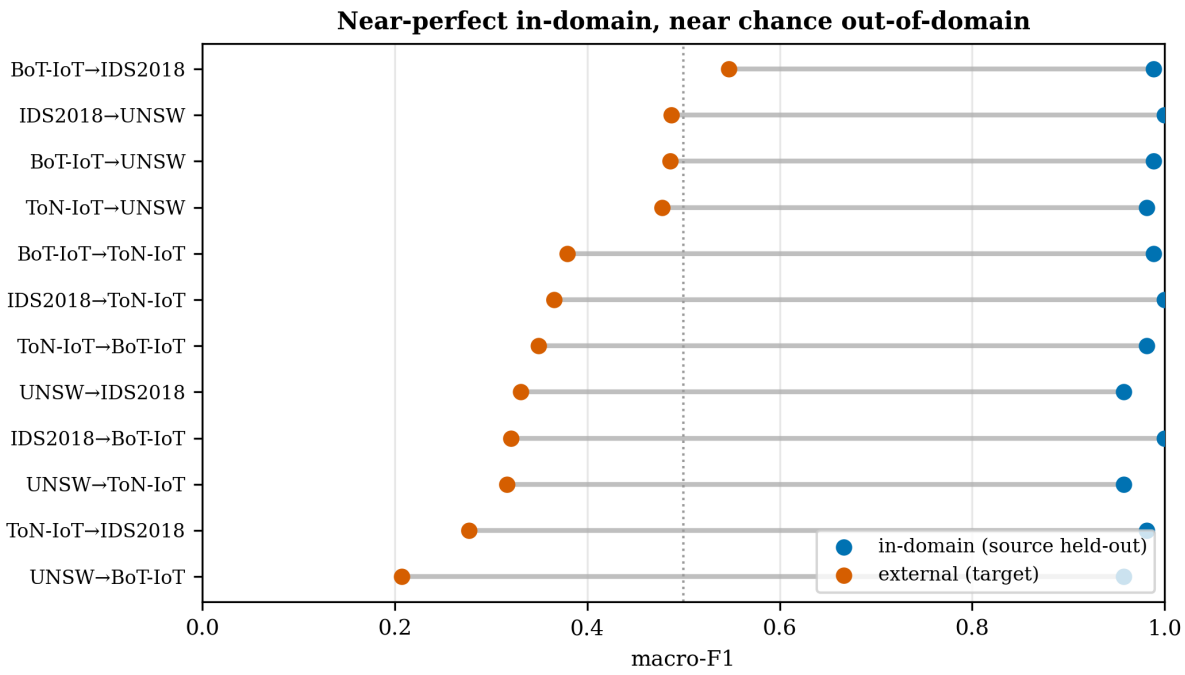


Figure 1: In-domain versus external macro-F1 for all 12 ordered pairs, LightGBM, mean over 5 seeds.

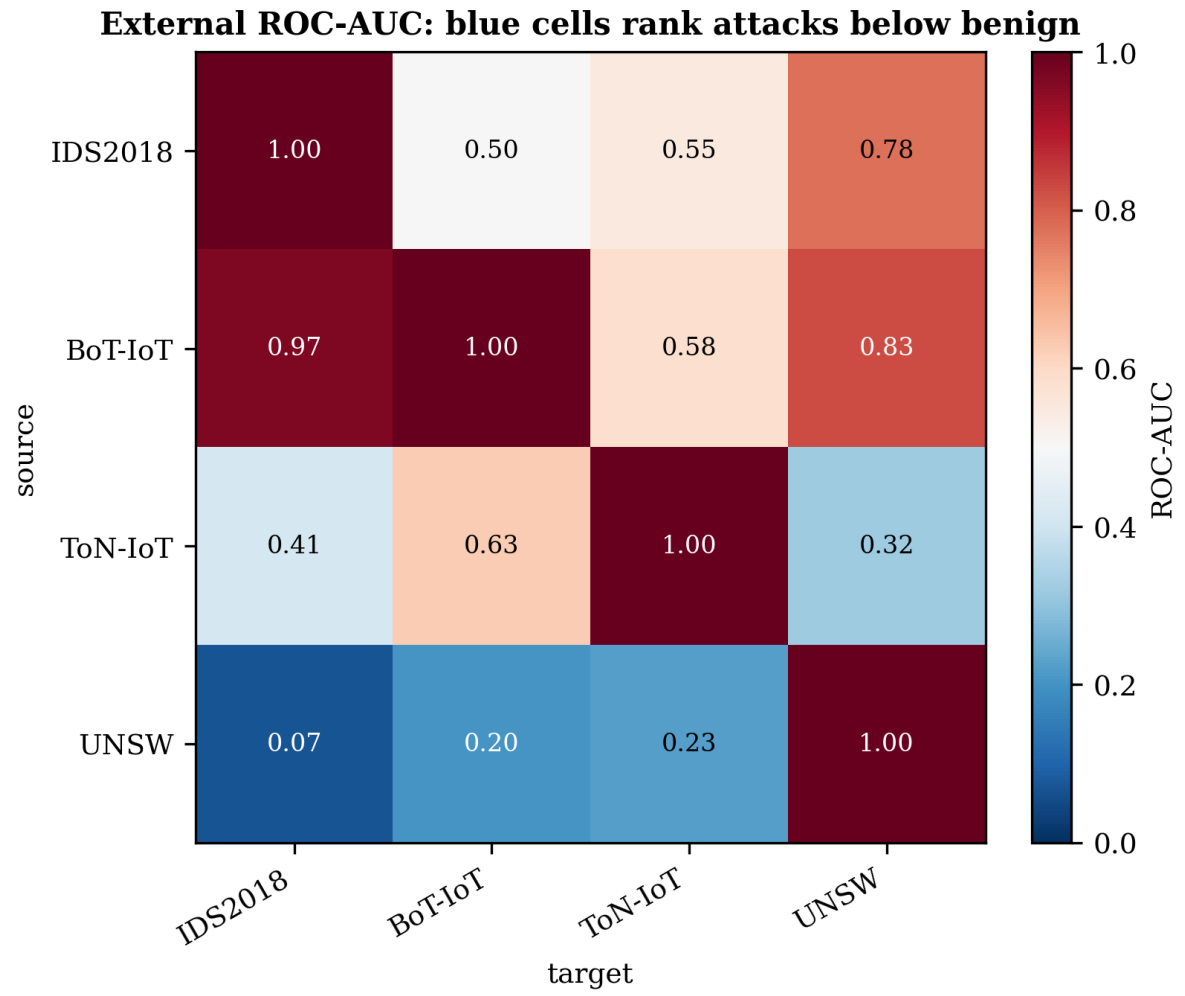


Figure 2: External ROC-AUC per ordered pair. Values below 0.5 indicate the ranking inverts in the target domain.

Table 3: Model-family robustness. External performance over all 12 ordered pairs and 5 seeds. Intervals are descriptive run-level bootstrap intervals over the 60 runs in each row; they are not pair-level inferential intervals, for the reason given below. The last column is the single worst run, not a pair mean.

model	n	mean external AUC	95% CI	below 0.5	mean macro-F1	mean MCC	lowest run
Logistic regression	60	0.5091	[0.4448, 0.5735]	41.7%	0.4789	+0.0590	0.0668
Random forest	60	0.5499	[0.4925, 0.6085]	40.0%	0.3665	−0.0596	0.1360
XGBoost	60	0.5184	[0.4537, 0.5840]	51.7%	0.3988	−0.0063	0.0339
LightGBM (primary)	60	0.5054	[0.4391, 0.5718]	46.7%	0.3788	−0.0459	0.0421

No family escapes. Mean external AUC lies between 0.5054 and 0.5499 for all four and the share of runs below chance between 40.0% and 51.7%; three of the four have negative mean MCC. Random forest transfers nominally best and still puts two runs in five below chance. Logistic regression carries the most weight here, because it is the one family that cannot represent interactions among features at all, and it inverts as readily as the ensembles do, with a worst run of 0.0668.

Averaging seeds within each pair shows that the failure is concentrated rather than diffuse.

Table 4: External ROC-AUC by ordered pair and model family. Mean over 5 seeds. Bold marks a ranking below chance.

source→target	LR	RF	XGB	LGBM	families below 0.5
BoT-IoT→IDS2018	0.9825	0.9742	0.9811	0.9656	0 of 4
BoT-IoT→ToN-IoT	0.6390	**0.4642**	**0.4569**	0.5833	2 of 4
BoT-IoT→UNSW-NB15	**0.3888**	0.7361	0.8191	0.8256	1 of 4
IDS2018→BoT-IoT	0.6717	0.6499	**0.4800**	0.5006	1 of 4
IDS2018→ToN-IoT	0.6884	**0.4967**	0.5805	0.5497	1 of 4
IDS2018→UNSW-NB15	0.6321	0.7791	0.7703	0.7750	0 of 4
ToN-IoT→BoT-IoT	0.6406	0.6064	0.7255	0.6279	0 of 4
ToN-IoT→IDS2018	0.6296	0.6848	**0.4904**	**0.4127**	2 of 4
ToN-IoT→UNSW-NB15	**0.0726**	0.5326	**0.3908**	**0.3241**	3 of 4
UNSW-NB15→BoT-IoT	**0.2334**	**0.2964**	**0.1910**	**0.2045**	4 of 4
UNSW-NB15→IDS2018	**0.3162**	**0.1729**	**0.0901**	**0.0697**	4 of 4
UNSW-NB15→ToN-IoT	**0.2139**	**0.2056**	**0.2449**	**0.2256**	4 of 4

Nine of twelve pairs fall below chance in at least one family; three do so in all four, and those three are exactly the transfers out of NF-UNSW-NB15, which §4.5 takes up separately. This is the more useful statement of the result. Inversion is not scattered noise that a different learner would average away: it attaches to particular source-target combinations and survives a change of hypothesis class, including a change to a model that can only add up single-feature contributions.

What counts as one observation. The counts above treat each (model, pair, seed) run as a unit, and those units are not independent — five seeds share a source dataset and four families share a split. Taking the ordered pair as the inferential unit instead, 5 of 12 pairs have a mean external AUC below chance, and the pair-level mean is 0.5054 [0.3589, 0.6529] over 12 pairs. That interval is wide, and it contains both substantial inversion and substantial above-chance transfer. We therefore do not claim that the population mean across conceivable network pairs is at or below chance; twelve pairs cannot support that. The claim this paper does defend is the one the interval leaves untouched: particular transfers invert systematically, repeatably across seeds, and in every model family tried. Throughout, run-level counts are reported as "of 240" and pair-level counts as "of 12", and the two are not interchangeable.

The rest of the protocol's metrics agree, and one of them is the operational headline. Table 2 commits

to reporting PR-AUC, Brier score and FPR at 95% TPR alongside the ranking and threshold measures; they are collected on every run and summarised here.

Table 5: Probabilistic and operating-point metrics. Means over runs; 80 in-domain (4 sources x 4 models x 5 seeds) and 240 external. ECE is reported separately in §4.8 from a dedicated calibration run at a different sample size.

model	regime	PR-AUC	Brier	FPR@95%TPR
Logistic regression	in-domain	0.8727	0.0582	0.1127
Logistic regression	external	0.4886	0.3448	0.7519
Random forest	in-domain	0.9847	0.0095	0.0077
Random forest	external	0.4721	0.3448	0.7261
XGBoost	in-domain	0.9897	0.0092	0.0068
XGBoost	external	0.4721	0.4148	0.7063
LightGBM (primary)	in-domain	0.9900	0.0086	0.0066
LightGBM (primary)	external	0.4734	0.4294	0.7142

PR-AUC falls to roughly chance for the balanced evaluation set in every family, so the collapse is not an artefact of macro-F1's treatment of the two classes. The Brier score rises by a factor between six and fifty. The number an operator would feel first is the last column: holding sensitivity at 95% of attacks, the false-positive rate on an unfamiliar network runs from 0.0066 to 0.7142 for the primary model. A detector that flagged two benign flows in a thousand at home flags seven in ten abroad, at the same recall.

4.3 Representation or operating point?

A collapsed macro-F1 admits two explanations with opposite remedies. The model may no longer separate the classes (a **ranking** failure, requiring retraining), or it may separate them well while the in-domain threshold has become meaningless (a **threshold** failure, requiring only recalibration).

Eleven of twelve pairs are ranking-dominated (Figure 3). The lone exception, NF-BoT-IoT $\rightarrow$ NF-CSE-CIC-IDS2018, retains ROC-AUC 0.9656, a ranking that transferred nearly intact, while losing 0.4217 macro-F1 to a stale operating point.

The optimistic reading available in the literature, that cross-domain collapse is largely a calibration artefact, is therefore true here for exactly one pair in twelve. §4.8 measures what that is worth.

The decomposition also gives an operator a decision rule, though not a free one. Ranking loss requires target labels to compute, so it cannot be measured before deployment. What §4.7 shows is that a label-free proxy for the same quantity exists and is moderately informative, which is why the two sections are ordered as they are: the decomposition establishes that the distinction matters, and the prediction section asks whether it can be made without labels.

4.4 Which features cause it? A pre-specified hypothesis the data does not support

Flow features divide into those describing *how much traffic moved and how fast* (throughput, duration, byte and packet volumes) and those describing *what the endpoints did* (protocol, TCP flags, TTLs, retransmissions). The first group's values are set partly by the capturing testbed's link capacity and generator configuration. The hypothesis this study was designed around is that these **capacity features** inflate in-domain accuracy while carrying testbed-associated structure, so removing them should recover external performance. It was committed to the project repository before the experiment ran; the commit recording it precedes the commit carrying the result by two hours and seventeen minutes. The repository was private at that time and made public on release, so this is time-stamped pre-specification in version history rather than public pre-registration, and we claim only the former.

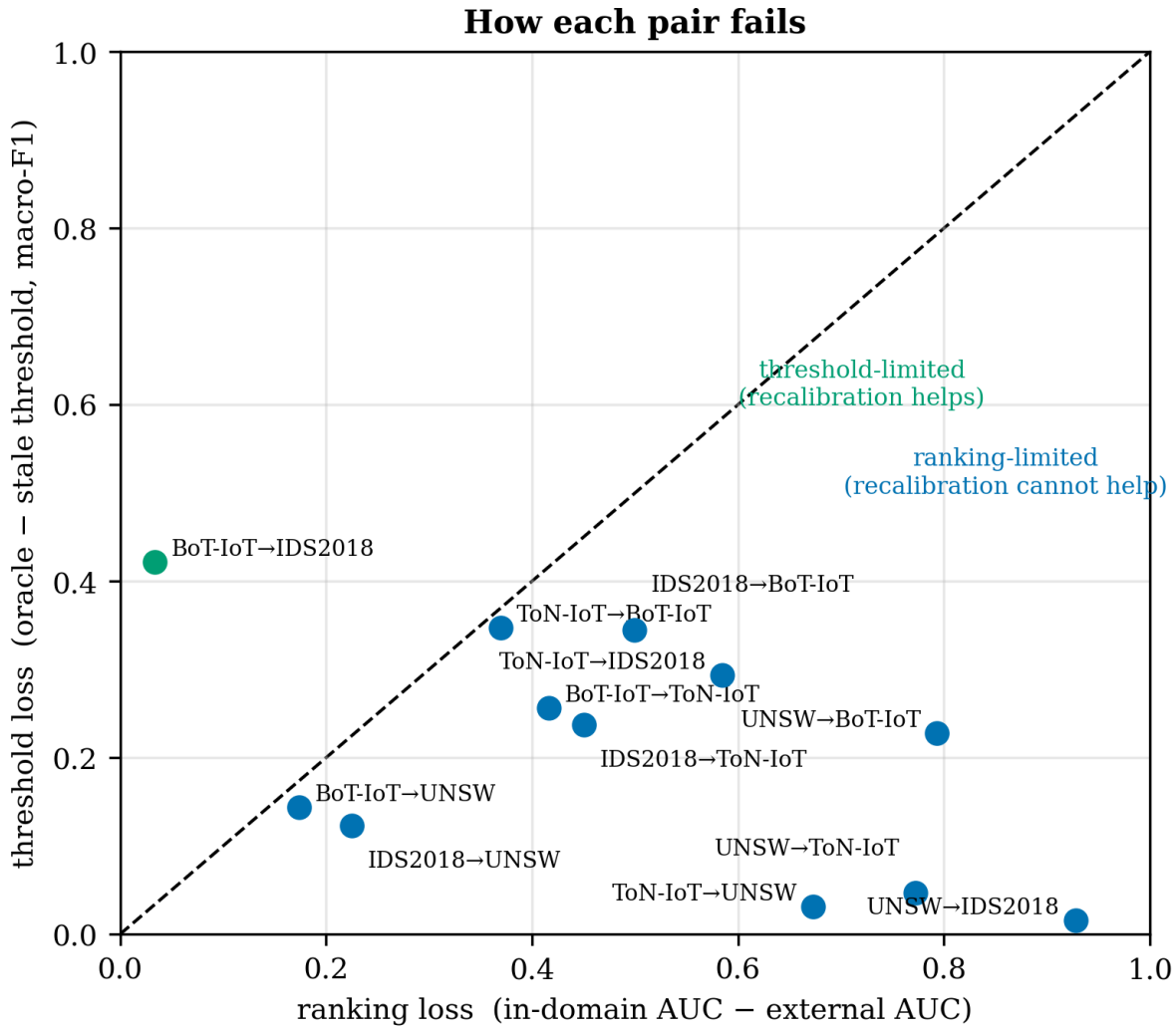


Figure 3: Ranking loss versus threshold loss per ordered pair. Points below the diagonal fail representationally; only one pair lies above it.

We assigned all 39 features to one of five semantic classes (20 capacity, 18 semantic, 1 held out) and trained under 17 feature sets. Each set is evaluated on the same (source, seed) combinations as the full-feature baseline, so differences are paired and the bootstrap resamples pairs.

Table 6: Capacity ablation, paired differences against the full 39-feature set.

feature set	k	external macro-F1	Δ vs `all`	95% CI	p	external AUC
`semantic_only`	18	0.3896	+0.0148	[−0.0077, +0.0341]	0.174	0.5394
`no_capacity`	19	0.3881	+0.0132	[−0.0084, +0.0320]	0.209	0.5386
`all`	**39**	**0.3748**	0	n/a	n/a	0.4931
`drop_rate_throughput`	35	0.3739	−0.0010	[−0.0033, +0.0013]	0.406	0.4882
`capacity_only`	20	0.3738	−0.0011	[−0.0293, +0.0314]	0.915	0.5221

Removing every capacity feature moves external macro-F1 by +0.0132, and the interval spans zero. The hypothesis is not supported. The last column adds something the pre-specified test could not see: on the ranking metric, capacity removal does move the needle, from 0.4931 to 0.5386, which is the difference between inverting and not. We flag this as an observation rather than a result — external AUC was not the pre-specified endpoint and no paired interval was computed for it — but it points the same way as §4.4's conclusion that feature choices act on the operating point and the ranking separately. We state that carefully: this is a failure to detect an effect, not a demonstration that none exists. The interval does not exclude effects up to roughly three points of macro-F1 in either direction, and with 12 source-seed pairs the study is not positioned to resolve differences of that size.

The coarse grouping does, however, hide one real sub-effect. Dropping the 13 size and volume features alone yields +0.0615 [+0.0278, +0.0991], $p < 0.001$, and dropping the three duration features yields +0.0114 [+0.0009, +0.0277], $p = 0.021$. Byte and packet counts do carry testbed-associated structure in the way the hypothesis anticipated. What fails is the aggregation: bundling them with the throughput features, which contribute nothing (−0.0010 [−0.0033, +0.0013]), dilutes the effect to non-significance. A semantic partition chosen a priori was the wrong instrument, and we report that rather than re-cutting the classes until something reached significance.

Smaller sets do transfer better, but not mainly because they are smaller. Ordering the same 17 sets by external performance puts the smallest on top (Figure 4).

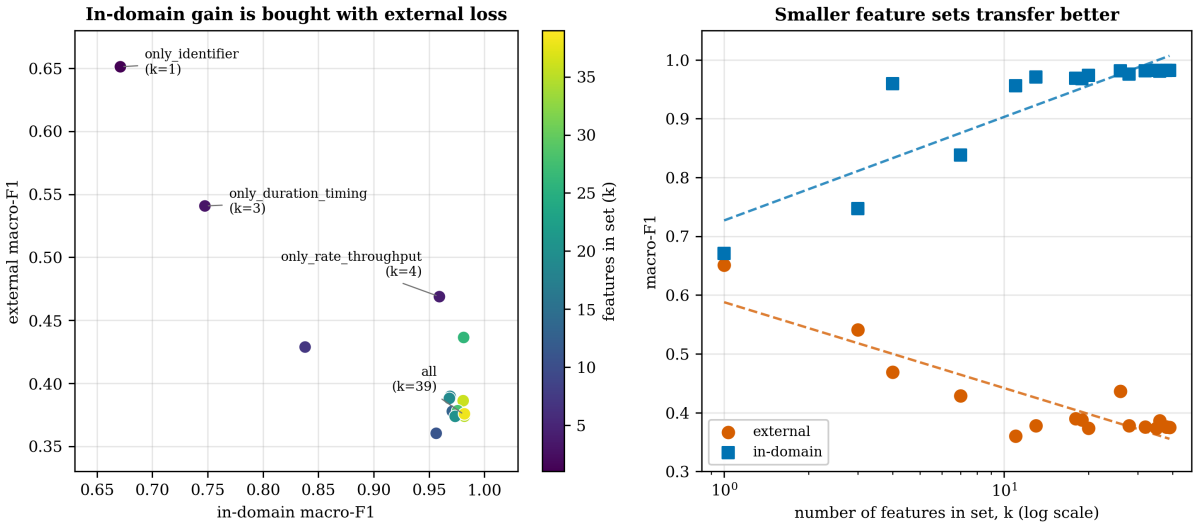


Figure 4: Feature-set size against in-domain and external macro-F1. The single-feature set attains the best external score of any subset tested.

Table 7: External performance by feature-set size.

feature set	k	in-domain	external macro-F1	Δ vs `all`	95% CI	external AUC
`only_identifier`	**1**	0.6711	**0.6512**	**+0.2763**	[+0.2230, +0.3366]	**0.7327**
`only_duration_timing`	3	0.7474	0.5407	+0.1659	[+0.0944, +0.2340]	0.5983
`only_rate_throughput`	4	0.9593	0.4688	+0.0939	[+0.0493, +0.1405]	0.6248
`only_flags_behaviour`	7	0.8380	0.4288	+0.0539	[+0.0371, +0.0705]	0.3759
`all`	39	0.9821	0.3748	0	n/a	0.4931

Every one of these differences is significant at $p < 0.001$. One feature out-generalises the entire standard set by 0.2763 macro-F1 [+0.2230, +0.3366].

That ordering cannot, on its own, tell dimensionality apart from feature identity: a one-bit DNS indicator and a four-feature throughput block differ in both. We therefore ran a control that varies size alone. For each k in {1, 2, 3, 5, 7, 10, 15, 20, 30, 39} we drew 8 subsets of size k uniformly at random without replacement from the 39 features, and trained on each of the 4 sources under 2 seeds at 25,000 rows per class, with splits, model and targets held fixed. At k = 39 only one subset exists, so that cell contributes 1 rather than 8, giving 584 runs.

Table 8: Random subsets at controlled size. Median over subsets, sources and seeds; 5th and 95th percentiles in brackets.

k	in-domain	external macro-F1	external AUC
1	0.7158	0.5046 [0.2835, 0.6973]	0.5889
3	0.9250	0.4214 [0.2950, 0.5881]	0.5336
10	0.9774	0.4021 [0.2928, 0.5195]	0.5766
20	0.9842	0.3993 [0.2786, 0.5111]	0.5898
39	0.9843	0.3997 [0.3017, 0.4406]	0.5843

Size matters, and less than the semantic ablation implied. Across random subsets external macro-F1 falls from 0.5046 at k=1 to 0.3997 at k=39, Spearman -0.290 [-0.369, -0.210]. That accounts for **+0.105** of the one-feature advantage. The remaining **+0.147** is identity: `DNS_QUERY_ID` alone reaches 0.6512 where a random single feature reaches 0.5046. Both components are measured inside the control, against its own median 39-feature baseline of 0.3997, whereas Table 7's +0.2763 is a mean paired difference against the full-feature runs at 0.3748. One baseline is a median over random subsets and the other a mean over the full-feature set, so the two components do not sum to +0.2763 and are not a partition of it. What survives either normalisation is that dimensionality is the smaller share: most of the one-feature advantage is the feature, not the count.

The size component is a threshold effect, not a representational one. Feature count correlates strongly with in-domain macro-F1 (Spearman +0.626 [+0.566, +0.679]) and negatively with external macro-F1 (-0.290), but is uncorrelated with external ROC-AUC (+0.004 [-0.081, +0.088]). Adding features sharpens separation at home and leaves the target ordering unchanged; what degrades is the usefulness of the operating point chosen against that sharper separation. This connects directly to §4.3 and to §3.6: perfect in-domain separability widens the tied-optimal threshold plateau and pushes the chosen threshold into a position that does not survive a change of network.

An earlier draft of this paper explained the ordering by supposing that a gradient-boosted ensemble given 39 correlated flow statistics finds high-order conjunctions encoding its own testbed, which a one-feature model cannot express. Two of our own results rule that out. Logistic regression inverts on the same pairs as the tree ensembles (§4.2) and cannot represent such conjunctions at all, and feature count is uncorrelated with external ranking quality, which is what a capacity explanation would predict it to govern.

We record the discarded explanation because the experiments that killed it are in this paper, and a reader who reaches the same intuition deserves to know it was tested.

A correction, stated where the claim depends on it. `DNS_QUERY_ID` was placed in its own class as a null control, on the assumption that a 16-bit DNS transaction ID is random and any accuracy from it would be memorisation by construction. That assumption is wrong.

Table 9: `DNS_QUERY_ID` is a protocol indicator, not an identifier.

dataset	P(attack | id = 0)	P(attack | id ≠ 0)
NF-CSE-CIC-IDS2018	0.741	0.000
NF-BoT-IoT	0.785	0.000
NF-UNSW-NB15	0.051	0.001
NF-ToN-IoT	0.528	0.427

The field is zero for every non-DNS flow, making it a one-bit ”this flow is DNS” indicator; in three of four datasets DNS traffic is essentially never attack-labelled. The ID value itself carries nothing: AUC on the raw value equals AUC on the zero/nonzero bit to four decimal places in all four datasets, and conditioning on nonzero leaves AUC at 0.48–0.49. The correct statement is therefore that a **one-bit DNS indicator out-generalises the 39-feature standard set**, and we read that as a finding about the benchmark rather than as a detector worth deploying. A field that separates the classes this well across three of four independently built testbeds, while carrying no information about attack behaviour, is a dataset-construction shortcut: DNS traffic was generated as background in these testbeds and was therefore almost never labelled as attack. Its in-domain score of 0.6711 is unusable in any case. What it establishes is that feature count and semantic quality are different things, and that the standard feature set contains at least one shortcut that transfers better than the behaviour the model is supposed to be learning.

4.5 Is the strongest inversion a weak-source artefact?

The three pairs that invert in every model family (§4.2) share a source, and that source retains only 3,075 attack flows after de-duplication at a 1.30% base rate. Source identity and source size are therefore confounded in the study’s most extreme result, and the obvious alternative explanation is that a detector trained on very few positives simply fails unpredictably.

We separate the two by holding size fixed and varying identity: every source is starved to the same 3,075 attack flows that NF-UNSW-NB15 actually has, and the transfer matrix is rerun over two model families and three seeds.

Table 10: Source-size control. External ROC-AUC, 144 transfers.

regime	source	n	mean external AUC	below chance
matched (3,075 attacks)	NF-UNSW-NB15	18	0.3889	72.2%
matched (3,075 attacks)	other three sources	54	0.6155	25.9%
full cap	NF-UNSW-NB15	18	0.2513	88.9%
full cap	other three sources	54	0.6147	20.4%

Starving the other three sources does not make them invert. Their mean external AUC moves from 0.6147 at the full cap to 0.6155 when reduced to NF-UNSW-NB15’s attack count, a difference of 0.0008. The inversion is therefore not explained by the number of positive training flows alone.

NF-UNSW-NB15 inverts under both regimes and remains 0.227 below the other sources at identical size. The explanation is a property of that network rather than of its sample.

One secondary effect is worth stating because it is not what we expected. NF-UNSW-NB15's external AUC is *higher* under the matched regime (0.3889) than under the full cap (0.2513). The two regimes differ for this source in class balance rather than attack count: capping at 3,075 balances it, whereas the ordinary cap leaves roughly eight benign flows per attack. A more balanced source therefore inverts less severely, which is consistent with the threshold reading in §4.4 but is a single observation and we do not build on it.

4.6 Do the preprocessing choices manufacture the result?

Three decisions in §3.3 and §3.4 could in principle produce the effect this paper reports rather than expose it. Each is ablated over four sources and two seeds, holding everything else fixed.

Table 11: Preprocessing sensitivity. Mean over four sources and two seeds. The primary configuration is marked.

configuration	in-domain macro-F1	external macro-F1	external ROC-AUC
de-duplicated (primary)	0.9857	0.3952	0.5229
duplicates retained	0.9968	0.4345	0.5476
contradictory groups excluded	0.9857	0.3952	0.5229
clipped, LightGBM (primary)	0.9857	0.3951	0.5245
unclipped, LightGBM	0.9857	0.3951	0.5245
clipped, logistic regression (primary)	0.9195	0.4834	0.5295
unclipped, logistic regression	0.9195	0.5057	0.5460

De-duplication changes what the study would have concluded. Retaining duplicates lifts in-domain macro-F1 by 0.0111 and external macro-F1 by 0.0392, which makes the argument in §3.3 concrete: a study that splits these captures without removing exact duplicates reports in-domain numbers it has not earned.

The more consequential effect is on the external side, and it does not run in one direction. Retaining duplicates raises NF-UNSW-NB15's external ROC-AUC from 0.1872 to 0.4486, because its 1,915,838 raw rows collapse to 203,641 distinct ones and the surviving copies are not distributed evenly across the decision surface. Severe inversion would have been reported as ordinary near-chance failure. On NF-BoT-IoT the same choice moves external ROC-AUC the other way, from 0.7858 down to 0.6126. Duplicate contamination does not simply inflate results; it adds source-dependent noise to the very quantity this paper estimates, and on the strongest inversion in the study it would have hidden the finding.

Contradictory groups do not bind here. Excluding them changes nothing at all: the two rows of Table 11 are identical because, under the DoS-only policy, there are no such groups to exclude. Zero of 8,889,592 distinct feature vectors in NF-CSE-CIC-IDS2018 and zero of 203,641 in NF-UNSW-NB15 carry both labels. The 1,940 and 858 groups counted in §3.3 all pair a benign flow with a non-DoS attack and disappear with those rows. The arbitrary-survivor concern that motivates the check is therefore inapplicable to this analysis, which is worth stating precisely because the check was run expecting the opposite.

Clipping cannot matter for the tree ensembles, and slightly hurts the linear model. For LightGBM the ablation is exactly zero on every metric, and this is a property of the model rather than a measurement: every split threshold is learned inside the source training range, so clamping an out-of-range value to that range cannot move a row across any split. We report it because a null that follows from the model class is evidence about the design, not about the data, and reporting it as though it were the latter would be misleading.

Logistic regression is where the choice can bind, and there it is mildly harmful: not clipping improves external macro-F1 by 0.0223 and external ROC-AUC by 0.0165. Clipping touches between 4,014 and

51,468 target rows per source-seed, so the intervention is not negligible in scope. The direction matters for the paper's honesty rather than its conclusion. For the tree-based primary model clipping is immaterial; for logistic regression the chosen policy modestly reduces external performance. Neither manufactures the collapse. Unclipped logistic regression still transfers at 0.5460 overall, and on NF-UNSW-NB15 it still inverts, at 0.4087.

None of the three decisions creates the effect. One of them, de-duplication, would have materially changed the conclusion had it been skipped, in the direction of understating what we found.

4.7 Can an operator predict the loss without labels?

An operator deploying a detector on their own network has traffic but no attack labels. We ask whether any label-free quantity predicts the loss, evaluated leave-one-pair-out, with all seeds of a held-out pair excluded together, against the baseline of predicting the mean gap (LOPO MAE 0.0827). A predictor that cannot beat "assume the average drop" is not useful whatever its in-sample correlation.

Table 12: Label-free predictors of the observed gap.

predictor	inspects	Spearman	95% CI	LOPO MAE	vs baseline
ATC estimated accuracy	model outputs	$-$**0.726**	[$-$0.836, $-$0.532]	**0.0629**	$-$**24%**
mean confidence drop	model outputs	+0.503	[+0.168, +0.738]	0.0671	$-$19%
max per-feature Wasserstein	features	+0.433	[+0.103, +0.720]	0.0678	$-$18%
summed Wasserstein	features	+0.149	[$-$0.197, +0.491]	0.0930	+12%
proxy A-distance	features	$-$0.119	[$-$0.442, +0.242]	0.0849	+3%
RBF-MMD	features	+0.031	[$-$0.306, +0.385]	0.0910	+10%

The result splits by what the measure inspects (Figure 5). Measures of distance between feature distributions fail: proxy A-distance, RBF-MMD and summed Wasserstein all have intervals spanning zero and all three are worse than the mean baseline out-of-sample. This deserves stating plainly, because proxy A-distance in particular is the theoretical motivation behind a large body of domain-adaptation work. In this benchmark, these three marginal feature-distribution measures carry almost no information about how much a detector will lose. The result is not paradoxical: these measures describe the shift in $P(X)$, whereas the loss is driven by the shift in $P(Y \mid X)$, and the two need not move together. A benchmark on which they did would be a benchmark on which covariate shift dominated concept shift, and these four testbeds are evidently not that.

Measures of the model's own behaviour work, moderately. ATC as defined by Garg et al. [2022] picks the confidence threshold at which the source's own fraction of below-threshold points equals its error rate, then reports the fraction of target points above that threshold as a label-free estimate of target accuracy. We do not treat that estimate as an estimator of macro-F1; we use it as a *predictor*, fitting a univariate linear map from it to the observed external score and scoring that map leave-one-pair-out, with all seeds of a held-out pair excluded together so no pair informs its own prediction. ATC reaches $\rho = -0.726$ against the gap and +0.756 [+0.585, +0.861] against external macro-F1, reducing leave-one-pair-out error by 24%. Twelve ordered pairs is a small sample, and a 24% error reduction supports triage, a decision about which deployments need labelled validation first, rather than a guarantee.

4.8 What actually fixes it

Calibration collapses by two orders of magnitude. Mean ECE is 0.0043 in-domain and 0.4372 externally. These detectors do not become hesitant on an unfamiliar network; they stay confident and become wrong, which is the worse failure for an operator setting an alert threshold.

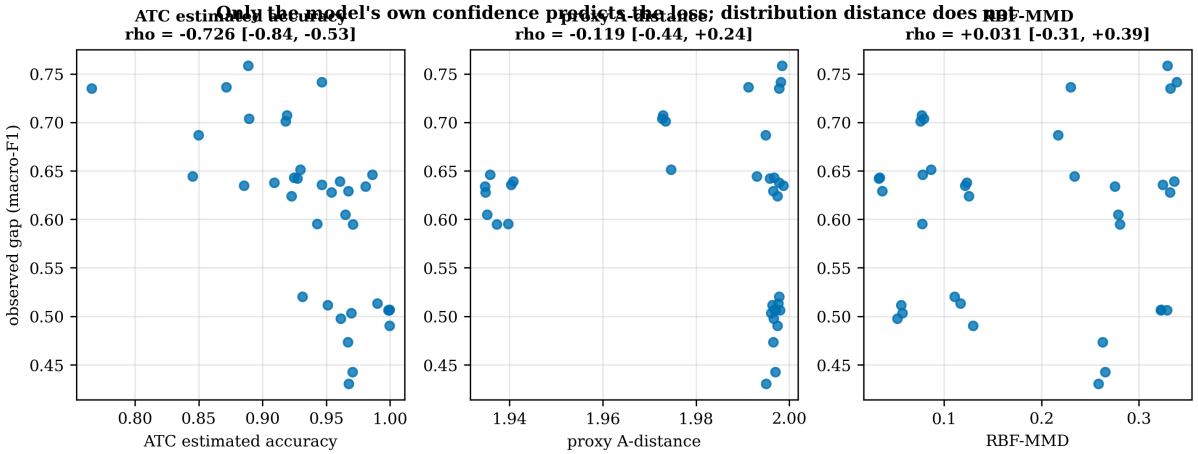


Figure 5: Observed generalization gap against three label-free predictors, with Spearman correlation and bootstrap CI.

Zero-label recalibration fixes the diagnosable minority, and harms others. A threshold read from the target's benign-traffic quantiles requires no attack labels. Its recovery, as a fraction of the headroom between the deployed threshold and the all-labels oracle, tracks external AUC almost exactly (Figure 6).

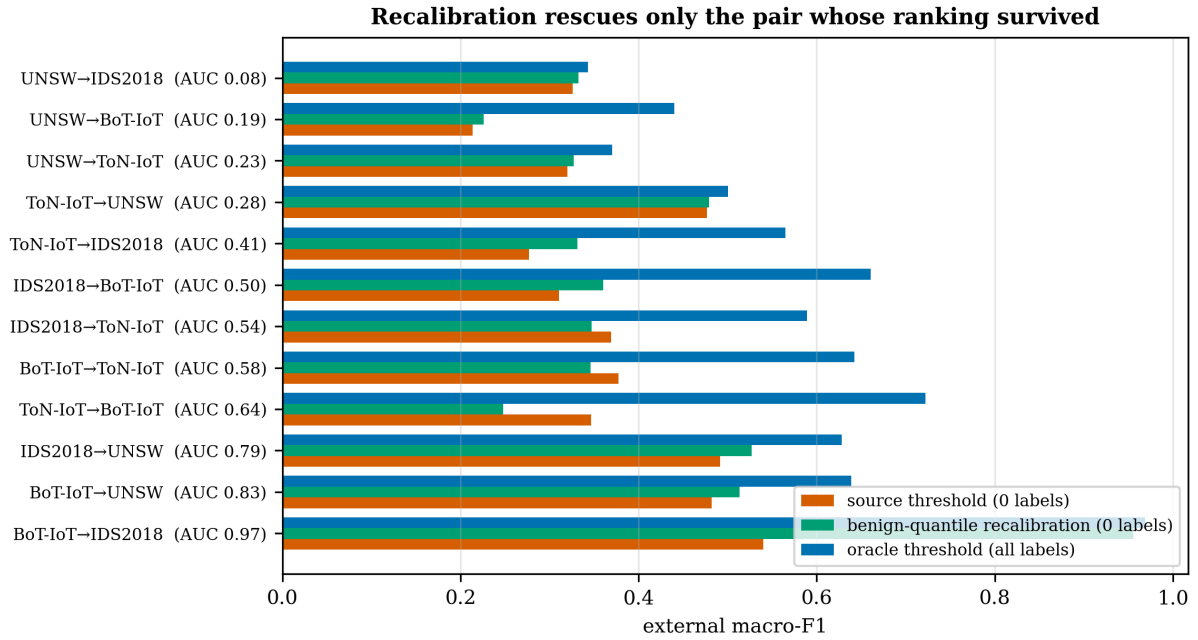


Figure 6: External macro-F1 under the deployed threshold, zero-label benign-quantile recalibration, and the all-labels oracle, ordered by external AUC.

It closes 97.0% of the one ranking-preserving pair and is actively harmful on three.

Pooled, it does not pay for itself. Scored on the held-out evaluation sets of §4.8.3, benign-quantile recalibration reaches median macro-F1 0.3482 against 0.3919 for the deployed source threshold, so averaged over all twelve pairs the procedure loses ground. The per-pair table is the useful reading and the pooled number is the warning attached to it: recalibration is worth doing when the ranking survived, and worth avoiding otherwise, which means an operator needs the diagnosis of §4.3 before choosing it rather than after. CORAL, a standard zero-label covariance alignment, recovers 0.0029 macro-F1 averaged over all pairs, which is indistinguishable from doing nothing. This bounds CORAL specifically; stronger representation-learning approaches to domain-invariant intrusion detection exist and are not evaluated

Table 13: Zero-label recalibration by pair.

pair	ext AUC	source thr	benign-quantile	oracle	headroom recovered
BoT-IoT→IDS2018	0.967	0.5399	**0.9558**	0.9688	**97.0%**
IDS2018→UNSW	0.792	0.4918	0.5268	0.6280	25.7%
BoT-IoT→UNSW	0.829	0.4820	0.5138	0.6386	20.3%
IDS2018→ToN-IoT	0.535	0.3690	0.3474	0.5890	−9.8%
BoT-IoT→ToN-IoT	0.583	0.3773	0.3465	0.6421	−11.7%
ToN-IoT→BoT-IoT	0.642	0.3469	0.2480	0.7220	**−26.4%**

here.

Ten balanced labelled flows outperformed the zero-label baselines evaluated here. Each budget k is drawn as k/2 benign and k/2 attack flows from the adaptation pool. That isolates the value of supervision from the difficulty of finding attacks, and it is an assumption: this is a controlled label-budget experiment, not a claim that ten flows sampled from live traffic would contain five attacks.

Table 14: Label budget, leakage-free. Median over 20 independent label draws per (pair, budget), 12 ordered pairs, 2 seeds; 5th and 95th percentiles in brackets.

k labelled target flows	re-tune threshold	retrain on source + k
0	0.3919 [0.2779, 0.5106]	0.3919 [0.2779, 0.5106]
10	0.5600 [0.2851, 0.9383]	**0.7253 [0.5604, 0.9540]**
50	0.6090 [0.3336, 0.9592]	0.8795 [0.7952, 0.9804]
100	0.6104 [0.3482, 0.9604]	0.9116 [0.8393, 0.9932]
500	0.6135 [0.3503, 0.9600]	0.9548 [0.8958, 0.9974]
1000	0.6142 [0.3620, 0.9609]	0.9574 [0.9228, 0.9986]

Each target is split once per (pair, seed) into a disjoint adaptation pool and evaluation set. Labels are drawn only from the pool and every score is computed only on the evaluation set, including the zero-label baselines, so the comparison is like for like. Draws are balanced, k/2 per class: at a 1.3% base rate a prevalence-proportional draw of ten flows usually contains no attack at all, which would measure the base rate rather than the value of supervision.

An earlier version of this experiment fine-tuned on k target rows and then scored a target set that still contained them, and tuned the threshold on rows it went on to score. We report the corrected design here. The direction of the error is worth stating: the contaminated run gave 0.7099 at k=10 against 0.7253 for the clean one, so the leak was depressing the estimate slightly rather than inflating it, and the conclusion is unchanged.

Ten balanced labelled flows lift median external macro-F1 from 0.3919 to 0.7253, a hundred reach 0.9116, and a thousand reach 0.9574 against an in-domain ceiling near 0.98 (Figure 7). Re-tuning the threshold on the same labels plateaus at roughly 0.61, and the reason is structural rather than empirical: changing a threshold cannot restore discrimination when the score ordering itself is reversed, because a threshold only chooses a cut point on an ordering it cannot change. Recovering the ordering requires changing the scoring function, which is what target labels buy.

The sharpest comparison is against a **target-optimal threshold oracle**: the best threshold obtainable on the evaluation set with unlimited target labels, which reaches 0.6158. It is an oracle rather than a zero-label method, since it consumes target labels without limit. Retraining on **ten** balanced labelled flows reaches 0.7253 and so exceeds it. Recalibration and supervision are not two points on one scale; beyond a certain point recalibration cannot buy what a few labels buy immediately.

Recovery is uneven across pairs, and unevenly in an informative direction. NF-UNSW-NB15 to NF-CSE-CIC-IDS2018, the most strongly inverted transfer in the study at ROC-AUC 0.070, reaches 0.9076 [0.8174, 0.9566] with ten balanced labels. That is what inversion predicts: the ordering still carries the information, with the sign reversed, and a small labelled sample is enough to supply the sign. The operational reading, at the scope the evidence supports, is that in this benchmark a small balanced labelled target sample was substantially more effective than the zero-label baselines tested. We evaluated one covariance-alignment method and one threshold heuristic, not the space of unsupervised domain adaptation, and the claim is limited accordingly.

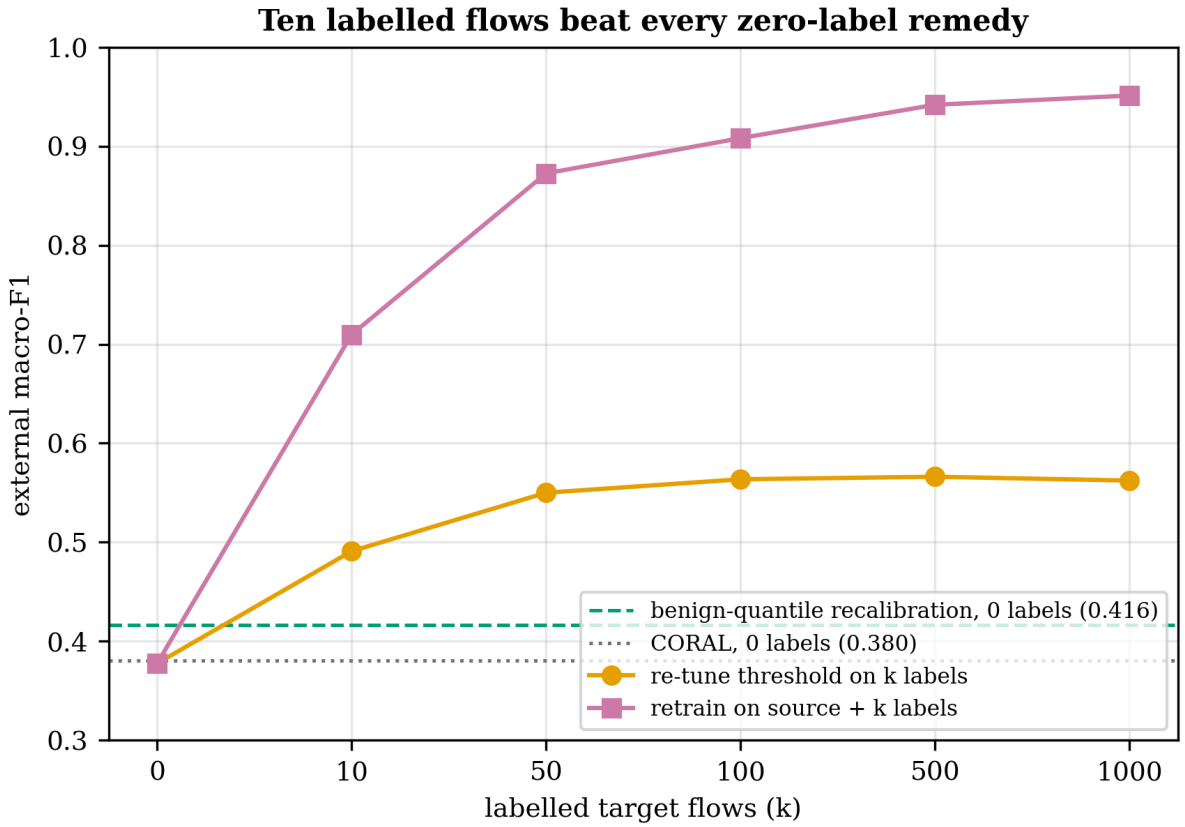


Figure 7: External macro-F1 against the number of labelled target flows, averaged over all ordered pairs and seeds.

4.9 Which attacks stop being detected

Table 15: Per-family recall under transfer.

attack family	in-domain	external	drop
volumetric	0.9834	**0.0535**	0.9299
low-rate application-layer	1.0000	0.2358	0.7642
benign (false-positive rate)	0.0118	0.0753	+0.0634

Volumetric floods, the largest and most conspicuous attacks and the class a detector is least expected to miss, transfer worst, retaining 5.35% recall. This follows directly from §4.2: a flood is defined by rates being high, and ”high” is a property of the capturing testbed. Low-rate application-layer attacks, defined by request patterns rather than volume, had roughly four times the external recall, an estimate limited by how few low-rate flows survive de-duplication.

4.10 Dataset pairing changes the observed failure mode

Repeating the study on the conventional CIC-DDoS2019 ↔ CICIDS2017 pairing over 76 shared CICFlowMeter features reproduces the gap: in-domain 0.9675–0.9995, external 0.3368–0.4986, mean paired gap 0.5848, significant in 10 of 10 pair-seeds. Neither the gap nor its size depends on the NetFlow v2 choice.

The decomposition does differ. Where the NetFlow quartet is ranking-dominated in 11 of 12 pairs, the CIC pairing is **threshold**-dominated in most configurations, and RandomForest retains ROC-AUC 0.9348 across it with ranking loss 0.0651 against threshold loss 0.5643.

CIC-DDoS2019 and CICIDS2017 share an institution, a testbed lineage and an extractor. Ranking largely survives that move, leaving an operating-point failure that recalibration can address. Moving between independently constructed testbeds destroys the ranking instead. The failure mode therefore depends strongly on the dataset pair. We have one same-lineage pairing against a cross-lineage quartet, so this is evidence consistent with a lineage effect rather than a demonstration of one: CIC-DDoS2019 and CICIDS2017 may be unusually similar for reasons other than shared provenance. Under either reading, a study confined to one institution's dataset family risks overstating how recoverable the problem is.

4.11 Correcting label errors makes transfer look worse

Table 16: Original versus corrected CICIDS2017, identical source model.

model	original	corrected	Δ macro-F1
LightGBM	0.4827	0.3375	−0.1453
XGBoost	0.4986	0.3391	−0.1595
RandomForest	0.4824	0.3381	−0.1443
LogisticRegression	0.4543	0.3287	−0.1256

Mean $|\Delta|$ is 0.1437, consistent across all four model families. For this source-target evaluation, correcting the target labels reduced measured macro-F1 by 0.1256 to 0.1595 across model families, so label error can materially inflate apparent transfer performance. Roughly a quarter of the external macro-F1 measured against the original labels does not survive correction, so any cross-dataset result using uncorrected CICIDS2017 as a target is optimistic by about this margin.

5 Discussion

The practical reading of these results is narrow and, we think, useful.

Validation on one network provides no empirical guarantee of behaviour on another, and the failure it will exhibit is not conservatism. It will keep issuing confident verdicts while its ranking is wrong or reversed, at an external ECE two orders of magnitude above its in-domain value. The three marginal feature-distribution measures evaluated here would not raise the alarm for these failures, because those distances did not track the loss. Monitoring the model's own confidence will, imperfectly.

The remedy hierarchy inverts the usual order of effort. Zero-label feature alignment by CORAL recovers 0.0029 macro-F1; a benign-quantile threshold helps substantially in the one regime where ranking survived and hurts in three others; ten balanced labelled flows recover 0.3323. For an operator, the cheapest useful action is also the most effective one: label ten flows from the network you actually run. The ordering holds for the zero-label methods we evaluated and is not a claim about unsupervised domain adaptation in general.

The finding that a single protocol bit out-generalises a 39-feature standard set should not be read as an endorsement of one-feature detectors, since its in-domain score of 0.6711 is unusable. It is a statement about what the other 38 features are doing, which is encoding a testbed at a 0.2764 external cost.

What this does not settle. The experiments do not identify which feature-label associations reverse sign; invariance across four hypothesis classes narrows the candidates without naming them. They do not show that unsupervised domain adaptation fails in general, only that one covariance-alignment method and one threshold heuristic did not recover the loss here. They do not show that drift monitoring is hopeless, only that three marginal divergence measures carried no usable signal across twelve pairs. And ten balanced labels are a controlled budget, not a deployment procedure. What the study does establish is a taxonomy of cross-network failure and the fact that which intervention helps depends on whether the target ranking survived.

Finally, §4.10 has a methodological consequence beyond this paper. Because the failure mode depends on whether the source and target share a lineage, the community's habit of evaluating transfer within a single institution's dataset family produces optimistic and qualitatively different conclusions from evaluation across families. Cross-institution evaluation should be the default.

6 Threats to validity

All four datasets are testbed captures. None is production traffic. This limits external validity in the usual way, but it sharpens the finding rather than weakening it: if detectors cannot transfer between four carefully constructed research testbeds, the case for transfer to a live network is not stronger.

De-duplication is a consequential choice. Removing exact duplicates in the post-socket-drop feature space reduces NF-BoT-IoT by 99.7%. A reader who considers those rows independent observations would obtain higher in-domain scores and a different gap. We report the counts so either position can be evaluated, and consider the alternative untenable: rows a model cannot distinguish cannot be independent test evidence for it.

The stratified cap alters base rates. Capping at 50,000 rows per class bounds cost but changes class balance, which moves precision-sensitive metrics. Base rates are reported throughout and operating-point metrics accompany F1 for this reason.

Non-DoS attacks are excluded from the primary analysis. Retaining them would change the base rate again; the policy is explicit and the counts moved are recorded, but the primary numbers describe a DoS/DDoS-versus-benign task.

Twelve ordered pairs is a small sample for §4.7. Leave-one-pair-out is the right protocol for the question, but a 24% error reduction estimated from twelve pairs supports triage, not a service-level guarantee.

NF-UNSW-NB15 is a small source. After de-duplication it retains 3,075 DoS flows at a 1.30% base rate, and it produces the study's most extreme inversion. The obvious confound is tested rather than conceded in §4.5: starving every other source to the same attack count leaves their transfer essentially unchanged (0.6147 to 0.6155), while NF-UNSW-NB15 remains 0.227 below them at identical size. Sample size does not account for the effect. Its class balance has a secondary influence that we note but do not interpret.

Low-rate attacks are underpowered in the NetFlow quartet. After de-duplication, NF-CSE-CIC-IDS2018 retains only 203 low-rate application-layer flows, so §4.9 rests partly on the CICFlowMeter secondary study.

Label error is bounded, not eliminated. CICIDS2017's defects are addressed against the corrected release; the NetFlow quartet has no comparably corrected counterpart, so residual label noise there is unmeasured.

The mechanism behind the inversion is not isolated. Our experiments rule two explanations out with-

out establishing a third. It is not a gradient-boosting artefact: logistic regression inverts on the same source-target pairs (§4.2) and cannot represent feature interactions at all. It is not model capacity expressed as feature count: external ranking quality is uncorrelated with the number of features (§4.4), which is the quantity a capacity account predicts it should track. What remains, and what we did not measure, is which individual feature-label associations reverse sign between networks. Invariance across four hypothesis classes is consistent with reversal in the low-order associations those classes have in common, but consistency is not identification, and we do not report it as a finding.

7 Conclusion

A flow-based DDoS detector moved to an independently constructed network does not become uncertain. It becomes confidently inverted: 108 of 240 run-level external evaluations score below chance, 5 of 12 ordered pairs do so once seeds are averaged, three of those invert in all four model families tried, and the worst pair reaches ROC-AUC 0.0697 — a ranking so consistently reversed that negating the score would yield 0.9303.

That distinction governs what can be done about it. Feature engineering along the obvious semantic axis does not help: the pre-specified capacity hypothesis is not supported (+0.0132 [−0.0084, +0.0320]), although size and volume features do matter individually (+0.0615 [+0.0278, +0.0991]). One protocol bit out-generalises the 39-feature standard by +0.2763 [+0.2230, +0.3366], but a control at matched size splits that advantage into +0.105 for dimensionality and +0.147 for the identity of that feature, a DNS indicator that separates the classes without describing attack behaviour. Feature count moves the operating point and leaves external ranking alone. Distribution-divergence monitoring does not predict the loss; model confidence partially does. Recalibration recovers almost everything in the one regime where ranking survived and is net harmful pooled across the others, and ten balanced labelled target flows beat the zero-label methods we evaluated and also exceed the target-optimal threshold oracle, which consumes unlimited target labels but can only move a cut point along an ordering it cannot repair.

Underneath all of it, these benchmarks are up to 99.7% duplicate, and correcting known label errors moves measured transfer *down* by 0.1437 macro-F1. Apparent transfer performance can therefore be moved substantially by data preparation and label quality before any model is chosen.

Data and code availability

All code, fixed seeds, configuration and result files are released. Every number in this paper is regenerated from `results/*.json` by the scripts that produced them; a verification script checks that each numeric literal in the manuscript traces to a stored result.

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